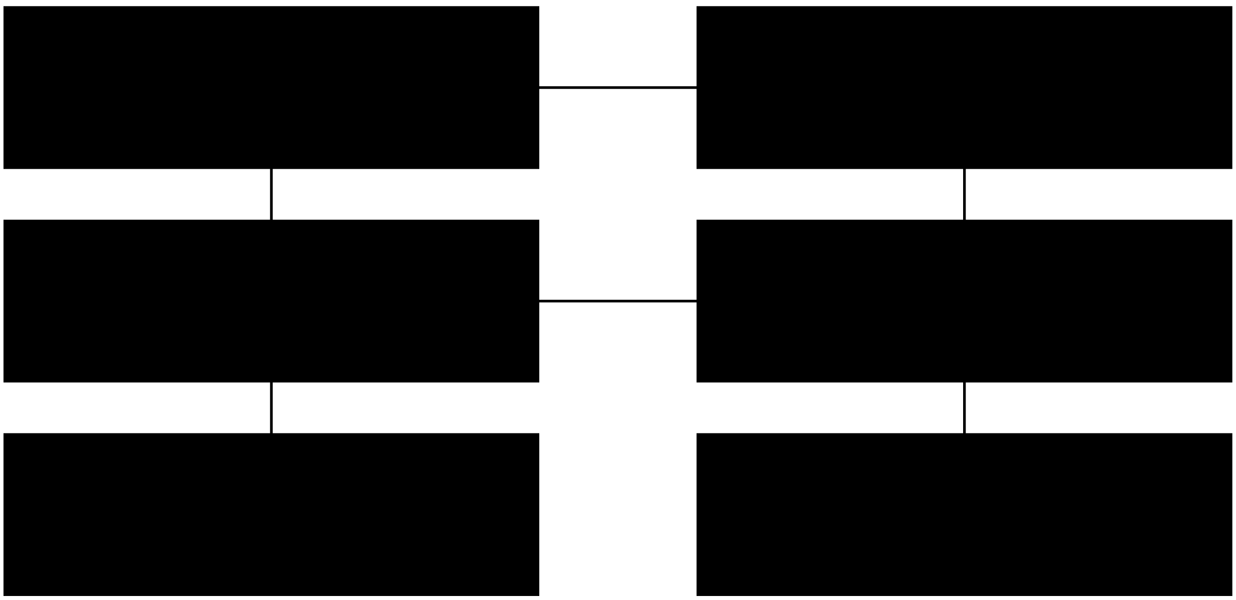
RCDEPS Ship Architecture — v1.0

Integrating the Unified Variable Model into propulsion, power, avionics, life support, and ISRU tooling.

1. System-of-Systems Overview

Mission intent: long-duration exploration and colonization with propellantless primary drive (RCDEPS), water-plasma fallback thrust, analog solar-thermal power, closed-loop life support, hydroponics, and ISRU/boring tools. All guidance computes with your Unified Variable Model (UVM): time-varying G, α, Λ; scalaron screening; torsion background.



2. Unified Variable Model (UVM) ↔ Avionics Map

UVM Quantity	Law / Callback	Consumers
$G_{eff}(t)$	$G_0 \cdot (1 + \lambda t + \beta \cdot \rho_{plasma}(t))$	Guidance, mass model, load cases
$\alpha_{eff}(t)$	$\alpha_0 \cdot (1 - \zeta B(t) - \chi_\alpha)$	EM calibration, sensor gains
$\Lambda_{eff}(t)$	$\Lambda_0 \cdot (1 + \varepsilon_{res} \sin \omega t)$	Low-rate cosmology tracker (monitor only)
$\lambda_{\phi,eff}(\rho)$	$75\ \mu m \cdot [1 + \rho/\rho_c]^{-1/2}$	Drive timing, resonance scan scheduling
τ_{total}	$\tau_0 + \kappa\sigma(r)$	Spin-sensitive metrology, RCDEPS phase bias

3. Propulsion Stack

Primary: RCDEPS pods (x8) with vectorable mounts; HV 0–20 kV, 0.1–10 Hz recursive modulation, $\beta/\phi/\tau$ control. Fallback: Water-plasma thrusters (x2) using metallic monolith grid packs or electrothermal steam; thrust for braking/rendezvous. Calibration: Photon-pressure pad for in-situ nN–mN reference ($F \approx 2P/c$).

Initial performance envelope (notional):

Subsystem	Power	Thrust	Isp / v_e	Notes
RCDEPS pod	0.5–2 kW	10■■→10■■ N	—	Coherence-scaled; vacuum only
Water-ion (grid)	50–300 W	0.5–5 mN	3–20 km/s	Needs neutralizer
Electrothermal steam	50–500 W	1–20 mN	0.5–2 km/s	Rugged; ISRU-ready
Photon pad	10–100 kW	0.07–0.7 mN	—	Reference/drag

4. Power Generation & Thermal

Analog-first: sun-tracking concentrators → steam generator → turbine/alternator → DC bus. Fail-safe loops, gravity-independent condensers, and radiator rails. PV & batteries as auxiliary.

No-fail features:

• Passive reliefs/orifices; • dual-loop isolation; • crowbar on DC link; • hardware current-limit to protect ECLSS.

5. Life Support, Hydroponics, Water

Closed-loop ECLSS: PEM electrolysis (O■), Sabatier (CH■ + H■O), solid amine CO■ capture. Hydroponics racks with LED spectra tuned to growth stages; water reclaim with catalytic recombiners; vacuum-compatible waste system (sealed, suction/valved, no free liquids).

6. GNC and Physics Monitor

Sensors: star tracker, IMU, sun sensors, LIDAR/radar for proximity. Physics: orthogonal optical cavities, atomic clock, one-way laser links on booms, photon pad. Software: EKF + UVM estimator → feeds $\beta/\phi/\tau$ planners and vectoring. Data: PPS-stamped telemetry with null runs beside active runs.

7. ISRU & Steam Borer

Steam-driven earth borer for subterranean habitats: helical cutter, jacketed with superheated steam ejectors; ejecta entrained and directed via nozzle ports. Use on-airless bodies with solar-thermal heat; water loop is closed and reclaimed.

8. Verification & Acceptance (Flight-Readiness Gates)

Gate	Test	Quantitative Acceptance
P1	Vacuum thrust (RCDEPS)	V^2 scaling, sign reversals, $\geq 5\sigma$ over null, repeatability
P2	Water-plasma module	mN thrust at set power; neutralization verified; erosion <1 μg/hr
M1	Physics monitor	Cavity beat stability; one-way TOF symmetry within spec; photon pad $F \approx 2P/c$ ±1%
E1	EMC/EMI	CISPR-11 lab compliance; no upset to avionics
S1	Safety interlocks	E-stop/crowbar <2 μs; bleeder <60 s to safe; enclosure interlocks
A1	Autopilot-in-the-loop	Closed-loop vectoring, MPC constraints, fault insertion pass

9. Top Risks & Mitigations

Risk	Mitigation
Thermal/ion artifacts masquerading as thrust	Vacuum-only runs; thermal dummies; cable symmetry; magnetic cleanliness
Grid erosion in water-ion mode	Rounded Mo/graphite grids; current-density limits; plume optics
Power brownout of ECLSS	Crowbar + hardware current-limit; segregated buses; load shedding
EMI into avionics	Faraday cages, fiber links, line filters, spectrum monitor

Steam overpressure	Reliefs/orifices; passive vents; dual containment

Appendix A — UVM Parameters (Snapshot)

$G\_eff(t)=G0(1+\lambda t+\beta\rho\_plasma(t))$; $\alpha\_eff(t)=\alpha0(1-\zeta B(t)-\chi\_\alpha)$; $\Lambda\_eff(t)=\Lambda0(1+\varepsilon\_res \sin \omega t)$; $\lambda\_\phi,eff(\rho)=75$ µm $[1+\rho/\rho\_c]$^{-1/2}; $\tau\_total=\tau0+\kappa\sigma(r)$.

Ship code reference: unified_model_v1_1.py and unified_params_v1_1.json (both included separately).